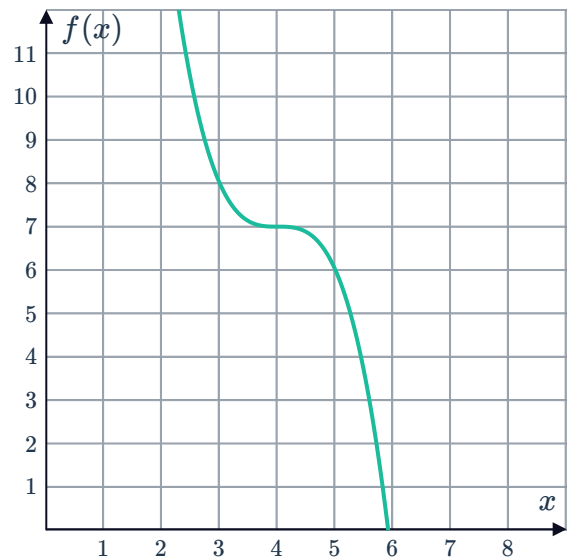


Functions and gradient functions



- 1 Consider the graph of the function $f(x) = -(x - 4)^3 + 7$:

- a State the x -value of the stationary point of $f(x)$.
- b State the domain where $f(x)$ is decreasing.



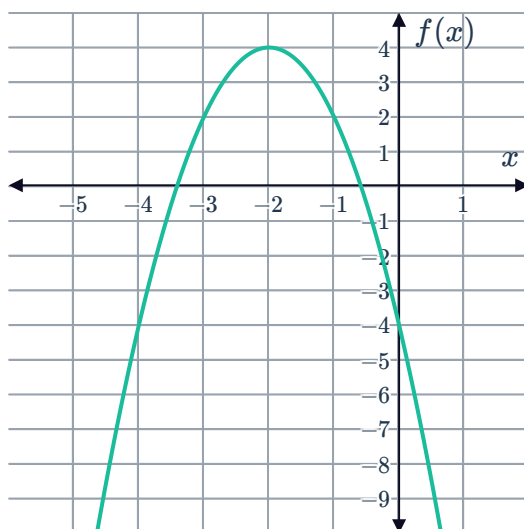
- 2 State whether the rate of change of the following functions is positive, negative, or zero for all values of x :

- a $f(x) = -8$
- b $f(x) = 4x - 1$
- c $f(x) = -x + 10$

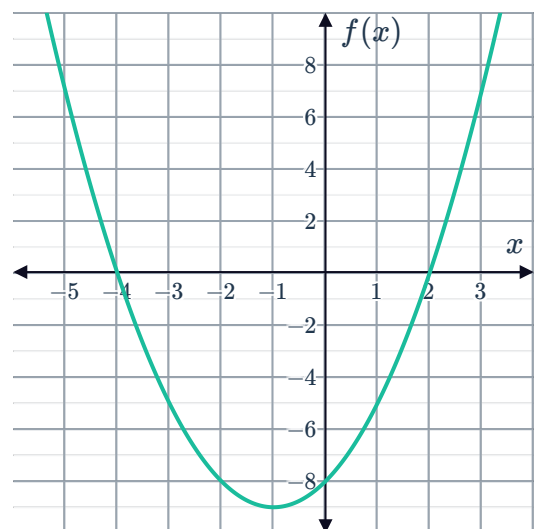
- 3 For each of the following functions:

- i State the x -value of the stationary point of $f(x)$.
- ii State the domain where $f(x)$ is increasing.
- iii State the domain where $f(x)$ is decreasing.

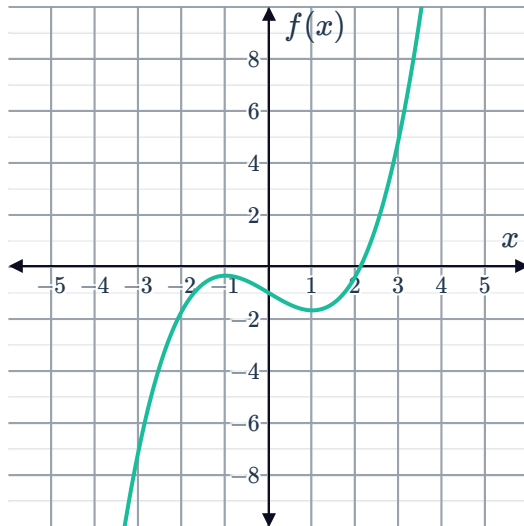
a $f(x) = -2(x + 2)^2 + 4$



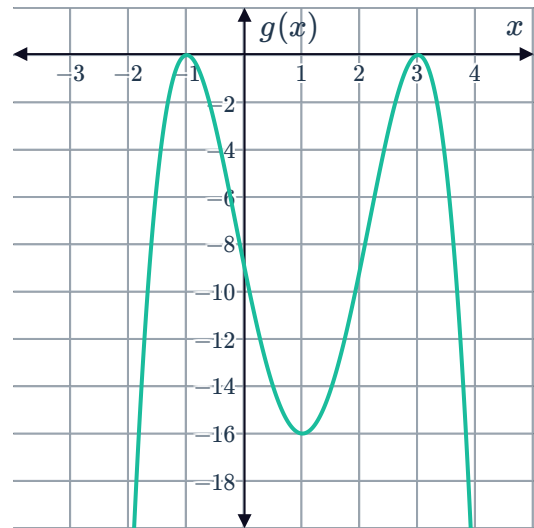
b $f(x) = (x + 4)(x - 2)$



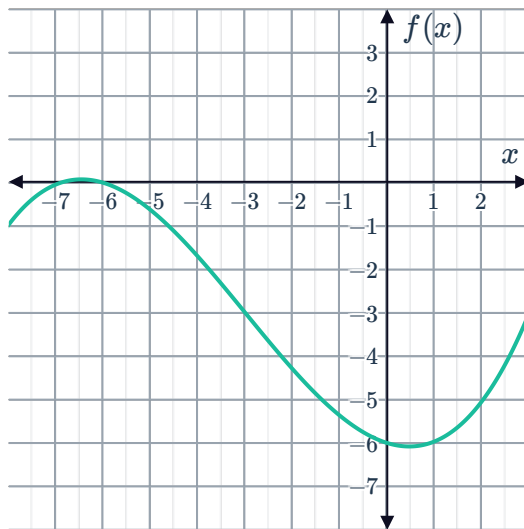
c $f(x) = \frac{x^3}{3} - x - 1$



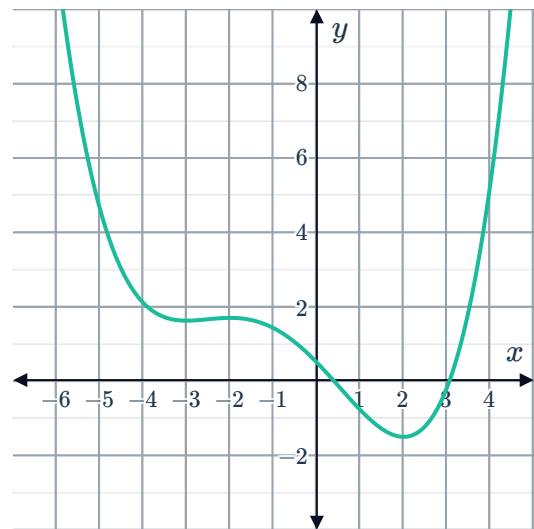
d $g(x) = -(x-3)^2(x+1)^2$



e $f(x)$



f $f(x)$

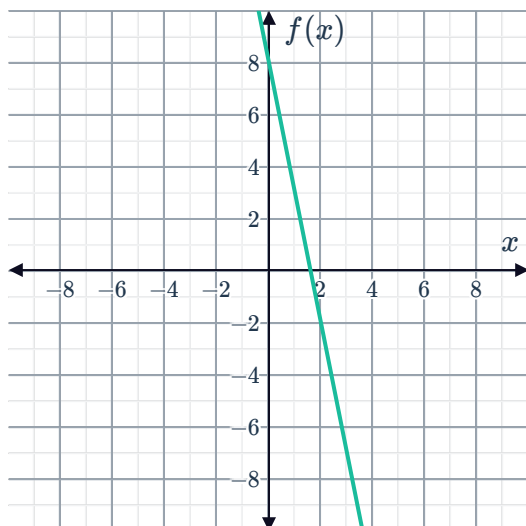


4 Consider the function $y = 4x - 3$.

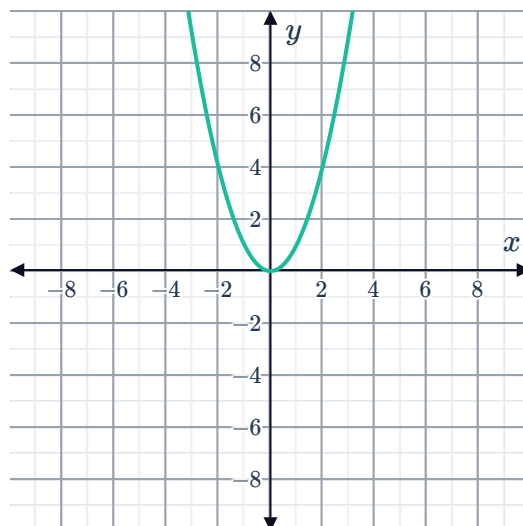
- Find the gradient function.
- Sketch the graph of the gradient function.

5 For each of the following functions, sketch  gradient function:

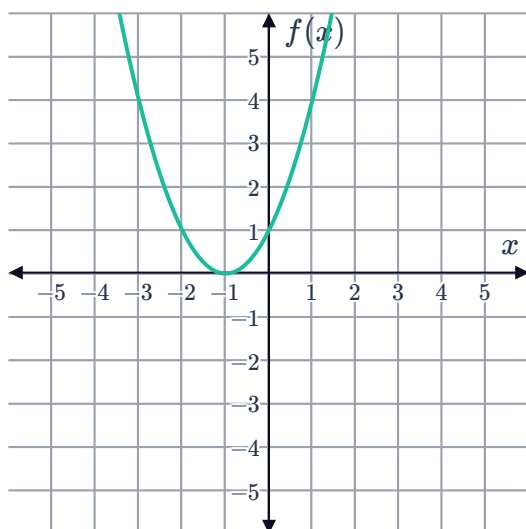
a $f(x) = -5x + 8$



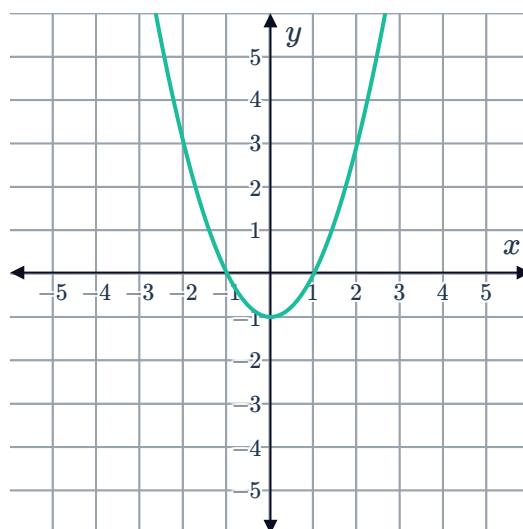
b $y = x^2$



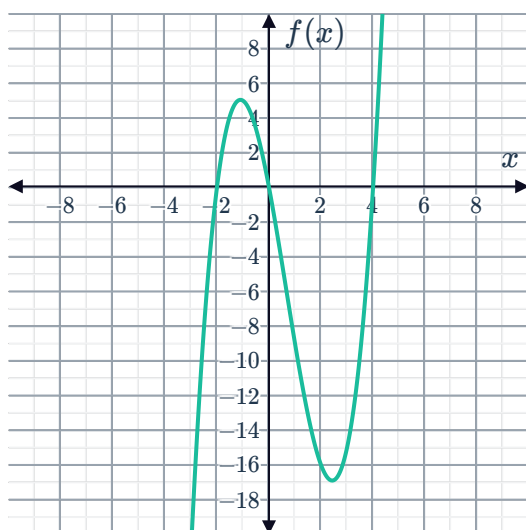
c $f(x) = (x + 1)^2$



d $y = x^2 - 1$



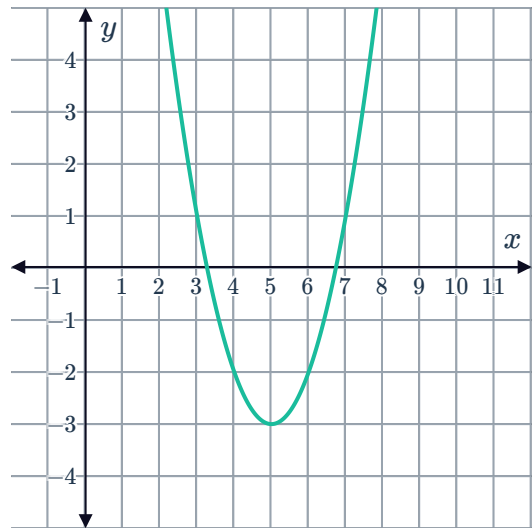
e $f(x) = x(x - 4)(x + 2)$



6 Consider the function $y = (x - 5)^2 - 3$ graphed below:

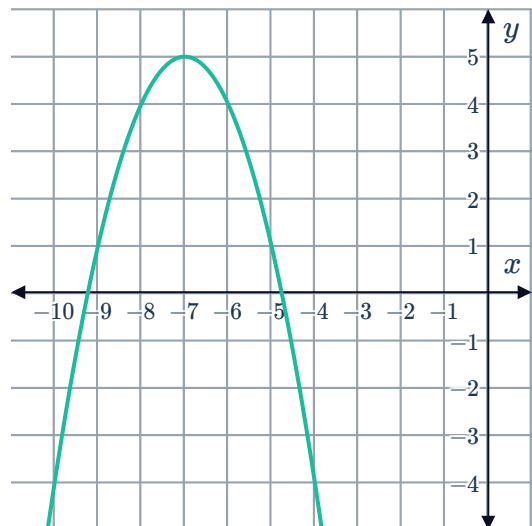


- a State the x -intercept of the gradient function.
- b For $x < 5$, state whether the values of the gradient function are above or below the x -axis.
- c For $x > 5$, state whether the values of the gradient function are above or below the x -axis.



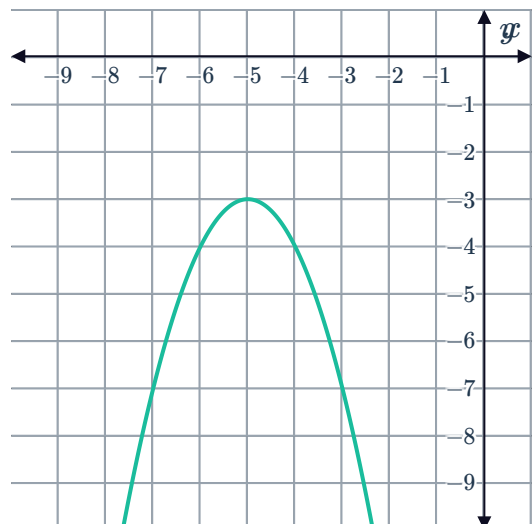
7 Consider the function $y = -(x + 7)^2 + 5$ graphed below:

- a State the x -intercept of the gradient function.
- b For $x < -7$, state whether the values of the gradient function are above or below the x -axis.
- c For $x > -7$, state whether the values of the gradient function are above or below the x -axis.



8 Consider the function $y = -x^2 - 10x - 28$ graphed below:

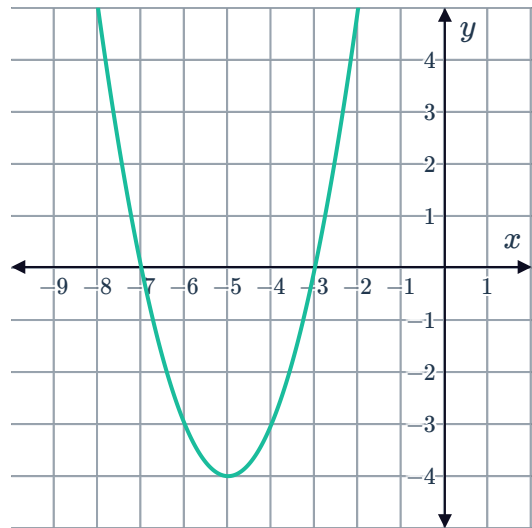
- a State the x -intercept of the gradient function.
- b For $x < -5$, state whether the values of the gradient function are above or below the x -axis.
- c For $x > -5$, state whether the values of the gradient function are above or below the x -axis.



9 Consider the function $y = x^2 + 10x + 21$ graphed below:

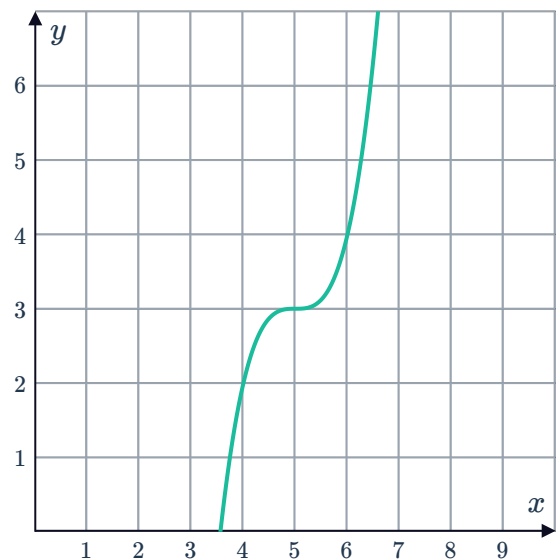


- a State the x -intercept of the gradient function.
- b For $x < -5$, state whether the values of the gradient function are above or below the x -axis.
- c For $x > -5$, state whether the values of the gradient function are above or below the x -axis.



10 Consider the function $y = (x - 5)^3 + 3$:

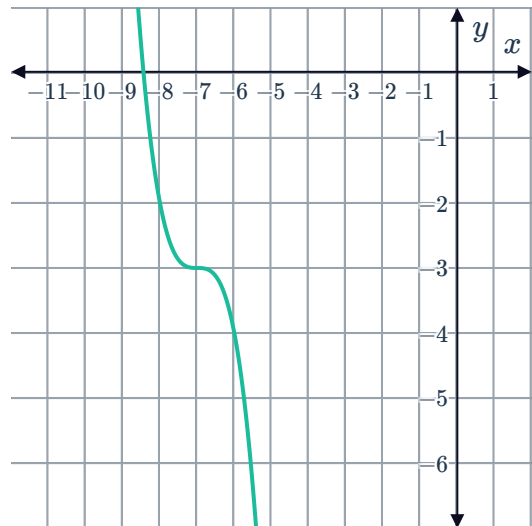
- a State the coordinates of the point of inflection.
- b State the gradient of the curve at this point.
- c What feature does the gradient function have at $x = 5$?
- d For $x < 5$, state whether the values of the gradient function are above or below the x -axis.
- e For $x > 5$, state whether the values of the gradient function are above or below the x -axis.
- f What type of point is on the gradient function at $x = 5$?



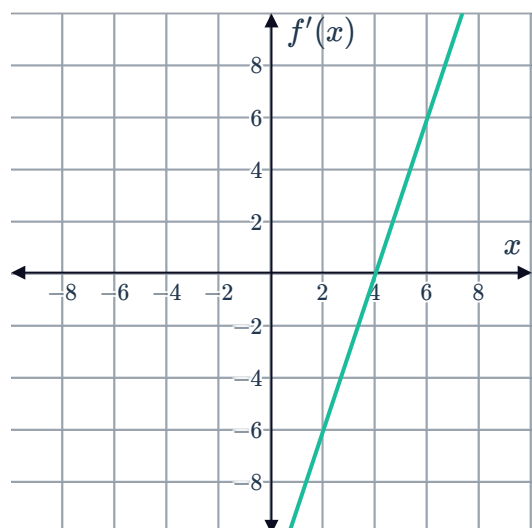
11 Consider the function $y = -(x + 7)^3 - 3$:



- State the coordinates of the point of inflection.
- State the gradient of the curve at this point.
- What feature does the gradient function have at $x = -7$?
- For $x < -7$, state whether the values of the gradient function are above or below the x -axis.
- For $x > -7$, state whether the values of the gradient function are above or below the x -axis.
- What type of point is on the gradient function at $x = -7$?



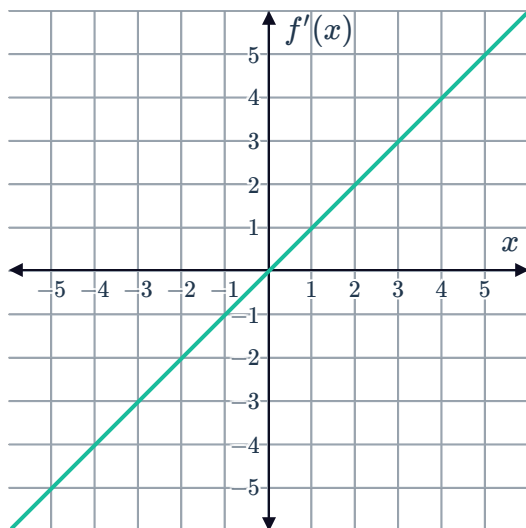
12 Consider the graph of the gradient function $f'(x)$:
What can be said about the graph of $f(x)$?



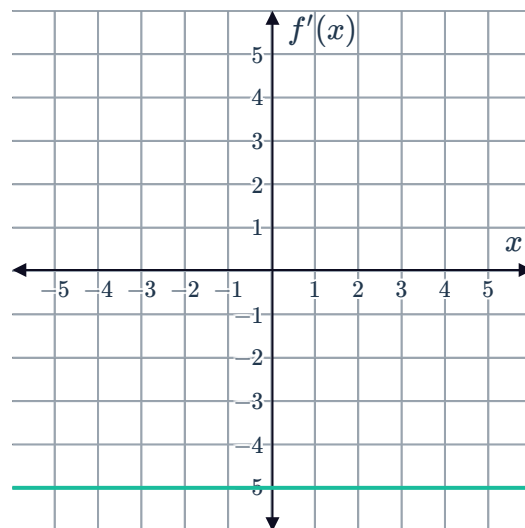
13 For each of the following gradient functions, sketch a graph of a possible original function:



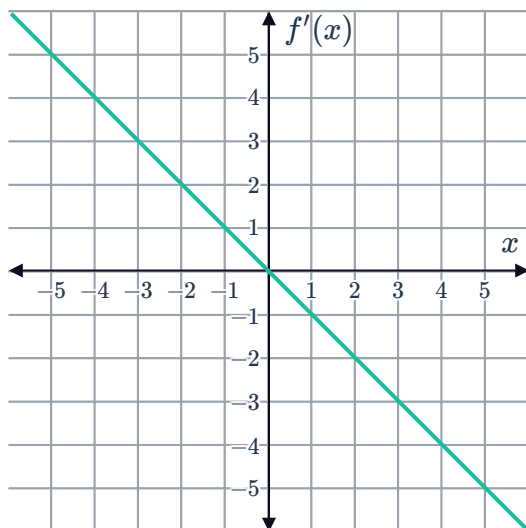
a



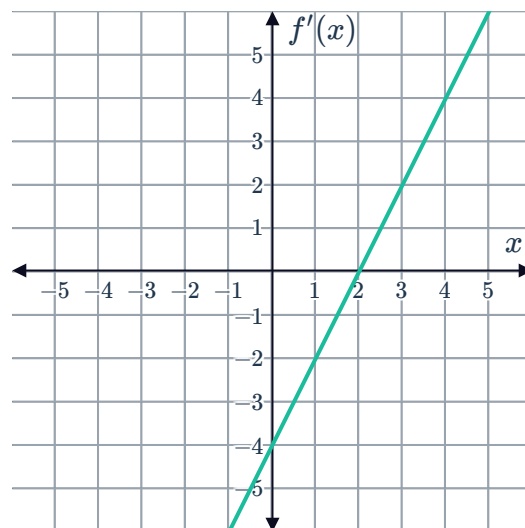
b



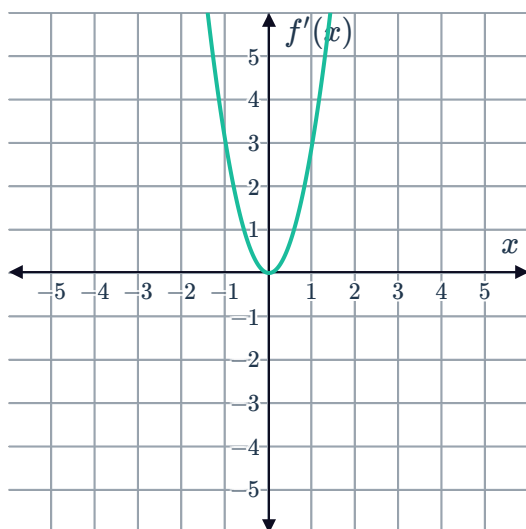
c



d



e





Types of stationary points

14 State the type of point that matches the following descriptions:

- a** A point where the curve changes from decreasing to increasing.
- b** A point where the curve changes from increasing to decreasing.
- c** A point where the tangent is horizontal and the concavity changes about the point.

15 For each of the following functions:

- i** Find the derivative.
- ii** Find the coordinates of any stationary points.
- iii** Classify each stationary point.

a $y = -6x^2 + 84x - 29$

b $y = x^3 - 21x^2 + 144x - 19$

c $f(x) = (x + 3)^2(x + 6)$

d $f(x) = (x + 5)^3 + 4$

e $f(x) = -\frac{x^3}{3} + \frac{13x^2}{2} - 30x + 10$

f $f(x) = (4x + 5)(x + 1)$

g $f(x) = 134 - 300x + 240x^2 - 64x^3$

h $f(x) = (x^2 - 9)^2 + 4$

16 Consider the parabola with equation $y = 5 + x - x^2$.

- a** Find the coordinates of the vertex of the parabola.
- b** State the gradient of the tangent to the parabola at the vertex.
- c** What type of stationary point is at the vertex of this parabola?

17 Consider the function $f(x) = x^2 + 4x + 9$.

- a** Find an equation for the gradient function $f'(x)$.
- b** State the interval in which the function is increasing.
- c** State the interval in which the function is decreasing.
- d** Find the coordinates of the stationary point.
- e** Classify the stationary point.

18 Consider the function $f(x) = 4x^3 + 5x^2 -$ 

- Find an equation for the gradient function $f'(x)$.
- Find the coordinates of the stationary points.
- Complete the table of values:

x	-2	$-\frac{5}{6}$	$-\frac{1}{2}$	0	1
$f'(x)$		0		0	

- Hence determine the:
 - Local minimum
 - Local maximum
- Is -4 the absolute minimum value of the function? Explain your answer.

19 Consider the function $f(x) = 3x^2 - 54x + 241$.

- Find $f'(x)$.
- Find the x -coordinate of the stationary point.
- Classify the stationary point.

Graphs of functions

20 Sketch the linear function for which $f(0) = 1$ and $f'(2) = 3$.

21 Sketch the quadratic function, $f(x)$, that satisfies the following conditions:

- | | |
|--|--|
| <p>a</p> <ul style="list-style-type: none"> $f(0) = -18$ $f(3) = 0$ $f(6) = 6$ $f'(6) = 0$ $f'(x) > 0$ for $x < 6$ | <p>b</p> <ul style="list-style-type: none"> $f(0) = 16$ $f(2) = 0$ $f(5) = -9$ $f'(5) = 0$ $f'(x) < 0$ for $x < 5$ |
| <p>c</p> <ul style="list-style-type: none"> $f(0) = 5$ $f(-2) = 0$ $f'(3) = 0$ $f'(x) > 0$ for $x < 3$ | <p>d</p> <ul style="list-style-type: none"> $f(0) = 10$ $f(-2) = 0$ $f(-6) = -8$ $f'(-6) = 0$ $f'(x) < 0$ for $x < -6$ |

22 Sketch the cubic function, $f(x)$, that satisfies the following conditions:



a

- $f(0) = 7$
- $f(-2) = 0$
- $f(-4) = -1$
- $f'(-4) = 0$
- $f'(x) > 0$ for $x < -4$
- $f'(x) > 0$ for $x > -4$

b

- $f'(2) = 0$
- $f'(-3) = 0$
- $f'(x) < 0$ for $-3 < x < 2$
- $f'(x) > 0$ elsewhere

23 Sketch the quartic function, $f(x)$, that satisfies the following conditions:

a

- $f'(-1) = 0$
- $f'(4) = 0$
- $f'(x) > 0$ for $x > 4$
- $f'(x) < 0$ elsewhere

b

- $f(0) = 0$
- $f'(0) = 0$
- $f'(2) = 0$
- $f'(-2) = 0$
- $f'(x) > 0$ for $x < -2, 0 < x < 2$
- $f'(x) < 0$ elsewhere

24 Consider the equation of the parabola $y = 3x^2 - 18x + 24$.

a Find the x -intercepts.

b Find the y -intercept.

c Find $\frac{dy}{dx}$.

d Find the stationary point.

e Classify the stationary point.

f Sketch the graph of the parabola.

25 For each of the following functions:

i Find the y -intercept.

ii Find the x -intercepts.

iii Find $f'(x)$.

iv Hence find the x -coordinates of the stationary points.

v Classify the stationary points.

vi Sketch the graph of the function.

a $f(x) = 9x^2 + 18x - 16$

b $f(x) = (4x + 5)^2(x - 1)$

c $f(x) = (2x - 1)^2(1 - x)$

d $f(x) = (x + 2)^3 - 1$

e $f(x) = x^3 + 11x^2 + 24x$

f $f(x) = (x^2 - 4)^2 + 4$

g $f(x) = 2(x - 1)^3 - 16$

h $f(x) = x^4 - 4x^3$

i $f(x) = x^3 + 5x^2$

j $f(x) = x^2(x - 3)^2$

26 Sketch the graph of the following function showing all stationary points:



a $y = 2x^3 - 12x^2 + 18x - 8$

b $y = 3x^5 - 20x^3 + 6$

c $y = 2x^3 + 3x^2 - 36x + 5$

d $y = 20 + 4x^3 - x^4$

e $y = -3x^4 + 16x^3 - 24x^2 + 30$

f $y = x^3 - 6x^2 - 15x + 7$